

Cryptocurrency Portfolio Analysis

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1. INTRODUCTION

This analysis is carried out on the 3 major crypto assets, Bitcoin (BTC), Ethereum (ETH) and Solana (SOL), to find out their risks and returns over the last 12 months. The aim is to analyze the performance of each asset, measure the risks and give allocation advice for various types of investors.

Over a 12-month period, the three holdings posted negative returns as digital assets faced a broad challenge. Yet, for each asset there are material differences in risk-adjusted performance, volatility and maximum drawdown. The dissimilarities have significant consequences for constructing portfolios and managing risks.

This analysis uses standard financial risk metrics. These include annualized volatility, maximum drawdown, value at risk, expected shortfall, along with the return based metrics. These also include total return, and Sharpe ratio.

2. PERFORMANCE OVERVIEW

Over the given period Solana suffered the largest losses down -45.1%. Ethereum was down 14.7% but managed to achieve the best Sharpe Ratio of the bunch at 0.04 the only number even close to breaking-even on a risk-adjusted basis.

Metric	BTC	ETH	SOL
Total Return (12M)	-21.0%	-14.7%	-45.1%
Average Monthly Return	-1.94%	-1.33%	-4.95%
Best Monthly Return	+19.8%	+27.6%	+38.2%
Worst Monthly Return	-23.1%	-28.4%	-34.7%
Sharpe Ratio (12M)	-0.45	0.04	-0.52

The average monthly return of -1.94 for Bitcoin was mild versus Solana (-4.95), but so negative that the Sharpe Ratios became, that none of these returns earned adequate compensation for the risk taken.

Ethereum's neutral Sharpe Ratio represents a significantly better return versus volatility ratio though it still finished the period negative.

During the 12-month data period, the three asset classes has a high spread between the strongest and weakest monthly return. The range of intra-period volatility the investor was exposed to is established by Solana's +38.2% to -34.7% over the last year..

3. RISK ANALYSIS

Risk metrics reveal significant divergence across the three assets. Bitcoin demonstrates the most contained risk profile, while Solana carries substantially higher volatility, drawdown, and tail-risk exposure. The table below presents the core risk indicators for each asset.

Risk Metric	BTC	ETH	SOL
Annualised Volatility	42.2%	67.2%	72.9%
Maximum Drawdown	-48.9%	-61.5%	-68.6%
Daily VaR (5%)	-3.80%	-5.69%	-6.03%
Expected Shortfall (5%)	-5.20%	-7.68%	-8.52%
Downside Risk Level	Medium	High	Very High
Risk Rank (1 = Safest)	1	2	3

Volatility

Bitcoin's annualised volatility of 42.2% is notably lower than ETH's 67.2% and SOL's 72.9%. While all three are high by conventional asset class standards, BTC's comparatively stable profile makes it the least demanding in terms of risk capacity required from an investor.

Maximum Drawdown

Solana experienced a peak-to-trough decline of -68.6% over the period — a figure that would require a subsequent gain of over 218% to fully recover. Ethereum's drawdown of -61.5% is similarly severe, while Bitcoin's -48.9% represents the least destructive outcome, though still a significant loss of capital from peak.

Value at Risk and Expected Shortfall

The 5% Daily Value at Risk (VaR) is a loss level which investors should not see surpassed on more than five per cent of the days. The performance of -3.80% of bitcoin is significantly better than that of sol at -6.03%. This means that on a typical bad day, sol holders lose 59% more than bitcoin holders.

An average of the losses that are incurred in the worst 5% of cases is what ES computes. According to the expected shortfall of SOL, whose value stands at -8.52% means the expected average loss on the absolute worst days is twice that of BTC. This is between -8.52% and -5.20%. It's the number we all care about.

Risk Ranking

Across all five risk metrics, BTC ranks first (safest), ETH second, and SOL third. This consistent ordering reflects a clear risk hierarchy among the three assets and should inform any allocation decision where downside protection is a consideration.

4. ASSET PROFILES

The following section provides an individual risk and suitability profile for each of the three assets, drawing on the performance and risk data presented above.

Asset	Risk Profile	Key Risk Factor	Stability Assessment
BTC	Moderate	Market cycles & macro shocks	Most stable of the three
ETH	High	Network upgrades & DeFi exposure	Moderately stable
SOL	Very High	Extreme volatility & network outages	Least stable

Bitcoin (BTC)

Of the three, Bitcoin has the best risk efficiency. With the lowest volatility and smallest drawdown to show for it, along with tail-risk metrics that are less onerous than its peers, it is the most defensible position in this portfolio. True, the return was a minus 21.0% over the period, but one can put that down to its sensitivity to macroeconomic shocks and the wider market cycle; a well known factor that has proven to be only temporary in the past. For the conservative crypto investor or anyone wanting a core, lower-risk allocation, BTC is the way to go.

Ethereum (ETH)

You could call Ethereum's performance moderately resilient when you put it up against Solana. It does have the finest risk-adjusted numbers of the group with a Sharpe Ratio of 0.04. But in absolute terms the risk is considerable: annualised volatility sits at 67.2% and we saw a maximum drawdown of -61.5%. Then there is the non-linear volatility that can come from the DeFi ecosystem at large or network upgrade cycles. ETH is for the balanced type of investor who is willing to put up with that kind of risk for the upside and exposure it offers.

Solana (SOL)

By any measure, Solana had the poorest risk-adjusted showing. A -45.1% return and a -68.6% max drawdown, not to mention an expected daily shortfall of -8.52%, put it squarely in the high-risk category. On top of the price action, SOL has a history of operational headaches like network outages that can make drawdowns worse when things get tough. This is an asset for aggressive investors with the stomach for sizeable losses and the time horizon to see them through.

5. ALLOCATION GUIDANCE

The allocation guidance below reflects each asset's risk profile and is structured by investor type. Given that all three assets declined over the period, the primary differentiator is not return but rather the degree of risk taken and the quality of risk-adjusted outcomes.

Investor Type	Preferred Asset	Rationale
Conservative	BTC	Lowest volatility and drawdown; most predictable risk profile. Recommended as a core holding within any crypto allocation.
Balanced	ETH	Best Sharpe Ratio in the group; accepts moderate-to-high risk in exchange for improved risk-adjusted return potential.
Aggressive	SOL	High return potential in bull markets; suitable only for investors with high risk tolerance and ability to absorb extreme drawdowns.

For investors holding all three assets, a BTC-weighted allocation would have provided the greatest capital preservation during this downturn. A suggested framework for a balanced portfolio might weight BTC most heavily, with a moderate ETH position and a smaller, tactical SOL allocation — sized to reflect the higher risk and potential reward profile of that asset.

It is worth noting that past performance and risk metrics should not be treated as predictive. The relative risk hierarchy observed over this 12-month window may not persist across different market regimes, and all positions should be reviewed in the context of an investor's overall portfolio, risk appetite, and investment horizon